BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Binary and Denary

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. How many bits are in a nibble and a byte?

2. What does an IDE's run-time environment let you do?
   A) Repair physical CPU faults
   B) Rearrange HDD fragments
   C) Manage network user accounts
   D) Run and test code

3. How do IDE error diagnostics help?

4. Two computers receive the same byte but show different characters. Why? What must match?

Today's objectives

Recall key facts about binary and denary. Work through a new example with correct technical language.

Key vocabulary - Read, cover, recall

Binary
Base-two number system using 0 and 1.

Denary
Base-ten number system.

Place value
The numerical weight of a digit's position.

Most significant bit
The leftmost bit with the greatest place value.

BLACK PEN: Write one definition in your own words.

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New Learning - Key ideas

Eight-bit place values

Read unsigned place values as 128, 64, 32, 16, 8, 4, 2, 1 from left to right.

Binary to denary

Add the place values in columns containing 1; ignore columns containing 0.

Denary to binary

Start with the largest place value that fits, subtract, then continue until the remainder is zero.

Check your answer

Eight-bit unsigned values range from 0 to 255; verify by adding the selected values.

I Do - Worked model

Convert unsigned binary 11001010 into denary. Show the place values used. [4 marks]

- Write place values
- The 1 bits are at 128, 64, 8 and 2.
- Add the selected values: 128 + 64 + 8 + 2.
- The result is 202 in denary. [4]

We Do - Guided practice

1. Convert unsigned 10110101 to denary. [4]

2. Convert 173 into an eight-bit unsigned binary number. [4]

3. Explain why 256 cannot be represented using eight-bit unsigned binary. [3]

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Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the power so the program prints the number of different 8-bit patterns.

```
patterns = 2^...
print(patterns)
```

You Do - Independent exam questions

Q2. Convert the unsigned binary value 01101001 to denary. Show your working. [3]

Q3. Convert denary 154 into an eight-bit unsigned binary number. Show your working. [4]

Q4. State the largest unsigned integer represented by eight bits and explain why. [2]

Exit Ticket - True or False?

An algorithm only gives reliable results when its inputs are handled as specified.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. A nibble has 4 bits and a byte has 8 bits.

2. D - Run and test code

3. They show errors and often their location, helping the programmer find the cause.

4. The computers use different character sets or encodings. To show the same letter, both need a matching encoding.

I Do - Explain the model

Write place values

The 1 bits are at 128, 64, 8 and 2.

Add the selected values: 128 + 64 + 8 + 2.

The result is 202 in denary. [4]

We Do - Model answers

1. 128 + 32 + 16 + 4 + 1 [2] = 181 [2].

2. 173 = 128 + 32 + 8 + 4 + 1 [2], so 10101101 [2].

3. Largest eight-bit unsigned value is 11111111 [1], equivalent to 255 [1]; 256 requires nine bits [1].

Code First - Answer

```
patterns = 2^8
print(patterns)
```

Expected output: 256

You Do - Self-mark

Q2: 64 + 32 + 8 + 1 [2] = 105 [1].

Q3: 154 = 128 + 16 + 8 + 2 [2], so 10011010 [2].

Q4: 255 [1], since all eight place-value bits set to 1 total 255 [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.